BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings illustrate example embodiments of the disclosed intraluminal compliance and distance profiling device. The drawings are schematic representations intended to support understanding of device structure and operation and are not necessarily to scale.

FIG. 4A illustrates an overall device configuration including an insertable section, distal sensing tip, compliance region, reference flange, and handle.

FIG. 4B illustrates an example distal sensing tip configured to measure distance using optical time■of■flight, LiDAR, ultrasound, or similar ranging methods.

FIG. 4C illustrates an embodiment including an inflatable compliance region configured to expand within the lumen for measurement purposes.

FIG. 4D illustrates an embodiment using pressure sensors along the insertable section to measure tissue interaction without inflation.

FIG. 4E illustrates alternative compliance configurations including localized cuffs, multiple sensing regions, or extended sensing coverage.

FIG. 4F illustrates a positional reference structure such as a flange or stop configured to provide repeatable insertion depth.

FIG. 4G illustrates a handle portion containing electronics, power components, communication modules, and optional pumping mechanisms.

FIG. 4H illustrates an example measurement process including insertion, distance acquisition, compliance measurement, and data recording.

FIG. 4I illustrates integration of device measurements into a broader computational system for generating user profiles and derived metrics.